Reviewer report: home-assistant-libs/pychromecast#1300: Expose multizone playback session info

Run `pychromecast-1300`, tested 2026-10-21 19:52 IST by ops/daily_run.py (route source-pair). Proof-of-Change commit `02a68eea38919fef2e1b537c3bf5a64bd7c771b5`. Report generated 2026-10-21 19:55 IST by reports/pr_report.py.

Baseline: **full** (test comparison + claim probes + coverage gate + claim flagging (differential probes shown, informational)).

NEEDS ATTENTION

Recommended action: **Reject** Confidence: **High**

The run is trustworthy and found a difference a person should look at. Every check passed and every claim cites a log line.

Advice only: Proof-of-Change never blocks a merge and never posts on GitHub.

Summary in plain English

- **What it says it does:** Expose multizone playback session info (home-assistant-libs/pychromecast#1300).
- **What we observed:** probe: 3 observation(s) need attention: c1.playback_session.initial_value does not differ although the claim says it does (C1); c1.playback_session.value_after_reset does not differ although the claim says it does (C1); c2.listener_callback_exists differs and no claim covers it; probe: claim(s) C1 proven; probe: 5 other observation(s) identical on both sides; tests: 1 test(s) ran on main, 6 on the PR; 5 new test(s) pass, 1 unchanged.
- **What you should do:** Do not approve as is. Read 'What changed in behavior' and decide with the PR author.

Verdict: needs_attention. probe: 3 observation(s) need attention: c1.playback_session.initial_value does not differ although the claim says it does (C1); c1.playback_session.value_after_reset does not differ although the claim says it does (C1); c2.listener_callback_exists differs and no claim covers it; probe: claim(s) C1 proven; probe: 5 other observation(s) identical on both sides; tests: 1 test(s) ran on main, 6 on the PR; 5 new test(s) pass, 1 unchanged

Model summary is advisory; the code verdict above is authoritative.

Agent swarm (advisory)

Role	Model	Status	Notes
judge	nvidia/Nemotron-3-Ultra-550b-a55b	ok	1 labels
helper	nvidia/Nemotron-3_5-Lightning	ok	c1.playback_session.initial_value: stable; ignore in future diffs if claim C1 context unchanged; c1.playback_session.value_after_reset: stable; ignore unless reset scenario activated; c2.listener_callback_exists: diff without claim; mark as unmonitored noise

Role	Model	Status	Notes
summary	nvidia/nemotron-3-super-120b-a12b	ok	model returned empty text; filled from code verdict/reason

judge/helper/summary are advisory; code owns the verdict

Demo helper

These notes are background from the console and the daily check. They do not change the verdict.

If setup fails, the helper retries the install up to 2 times. If a test still can't be verified because something is missing, it installs what's missing and reruns that test once.

- Setup check: GitHub token: present.
- Setup check: Nebius token: present.
- Setup check: free memory 6.4 GB (of 15.6 GB).
- Setup check: free disk 98.1 GB.
- Demo helper: git is installed.
- Demo helper: python3 is installed.
- Demo helper: uv is installed.
- Looked up with Tavily: 5 results
- Query: pychromecast python setup install instructions https://github.com/home-assistant-libs/pychromecast site:github.com/home-assistant-libs/pychromecast
- Query: pychromecast python documentation install https://github.com/home-assistant-libs/pychromecast site:github.com/home-assistant-libs/pychromecast
- Query: pychromecast python install requirements https://pypi.org/project/pychromecast/ https://github.com/home-assistant-libs/pychromecast site:pypi.org
- Source: PyChromecast · PyPI — https://pypi.org/project/pychromecast/
- Source: GitHub - home-assistant-libs/pychromecast: Library for Python 3 to communicate with the Google Chromecast. · GitHub — https://github.com/home-assistant-libs/pychromecast
- Source: Extra dependencies on raspbian Jessie (raspberry pi) · Issue #132 · home-assistant-libs/pychromecast · GitHub — https://github.com/home-assistant-libs/pychromecast/issues/132
- Source: pychromecast/pychromecast/socket_client.py at master · home-assistant-libs/pychromecast · GitHub — https://github.com/home-assistant-libs/pychromecast/blob/master/pychromecast/socket_client.py
- Source: "[WinError 10049] The requested address is not valid in its context" · Issue #483 · home-assistant-libs/pychromecast · GitHub — https://github.com/home-assistant-libs/pychromecast/issues/483
- Setup plan: the same packages on main and pr.
- Setup plan: protobuf is noted, installed with the project.
- Setup plan: zeroconf is noted, installed with the project.
- Setup plan: casttube is noted, installed with the project.
- Setup plan: install beautifulsoup4==4.15.0 (listed in requirements-test.txt) on main and pr.
- Setup plan: install ruff==0.17.1 (listed in requirements-test.txt) on main and pr.
- Setup plan: install mypy==2.4.0 (listed in requirements-test.txt) on main and pr.

- Setup plan: install PlexAPI==4.17.2 (listed in requirements-test.txt) on main and pr.
- Setup plan: install pylint==4.1.3 (listed in requirements-test.txt) on main and pr.
- Setup plan: install pytest==9.1.1 (listed in requirements-test.txt) on main and pr.
- Setup plan: install rstcheck==6.3.0 (listed in requirements-test.txt) on main and pr.
- Setup plan: install types-beautifulsoup4==4.12.0.20250516 (listed in requirements-test.txt) on main and pr.
- Setup plan: install types-html5lib==1.1.11.20260518 (listed in requirements-test.txt) on main and pr.
- Setup plan: install types-protobuf==7.35.1.20260906 (listed in requirements-test.txt) on main and pr.
- Setup plan: install types-requests==2.33.0.20261006 (listed in requirements-test.txt) on main and pr.
- Setup plan: install yle-dl==20240927 (listed in requirements-test.txt) on main and pr.
- Setup plan: Python 3.12: Python 3.12 already meets >=3.11.0
- Installed beautifulsoup4 4.15.0 from requirements-test.txt (listed in requirements-test.txt) on main and pr.
- Installed ruff 0.17.1 from requirements-test.txt (listed in requirements-test.txt) on main and pr.
- Installed mypy 2.4.0 from requirements-test.txt (listed in requirements-test.txt) on main and pr.
- Installed PlexAPI 4.17.2 from requirements-test.txt (listed in requirements-test.txt) on main and pr.
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- Installed types-requests 2.33.0.20261006 from requirements-test.txt (listed in requirements-test.txt) on main and pr.
- Installed yle-dl 20240927 from requirements-test.txt (listed in requirements-test.txt) on main and pr.

Background (related public bug reports, not used for the verdict)

1. "Issues · home-assistant-libs/pychromecast · GitHub" https://github.com/home-assistant-libs/pychromecast/issues "home-assistant-libs/pychromecast#1059 · opened on Apr 28, 2025 ### Fields in `media.metadata` (artist, title etc) never get cleared in `MediaStatus` and can report wrong/stale info. home-assistant-libs/pychromecast#1018 · opened on Feb 3, 2..." 2. "GitHub - home-assistant-libs/pychromecast: Library for Python 3 to communicate with the Google Chromecast. · GitHub" https://github.com/home-assistant-libs/pychromecast "namespaces ='urn:x-cast:com.google.cast.player.message''urn:x-cast:com.google.cast.media' session_id ='CCA39713-9A4F-34A6-A8BF-5D97BE7ECA5C' transport_id ='web-9' status_text = '' >> mc = cast media_controller >> mc play_media' >> mc block_..." 3. "[solved] Pychromecast not connecting to devices (zeroconf issue) - cast - Home Assistant Community" h...

Claims vs reality

#	Claim (source)	Status	What we observed	Log line
C2	Existing behavior is kept: tests that pass on main still pass (implied by every PR)	proven	no test that passes on main fails on the PR (1 unchanged, 3 runs per side)	pytest-pr-1.log:7 «6 passed in 0.97s»
P-C1	The PR parses the playbackSession field from MULTIZONE_STATUS messages and exposes it as MultizoneController.playback_session, which is None when no session is attached and cleared on reset_members(). (PR description)	not observed	probe model-probe-c1, c1.playback_session.initial_value: same on both sides: `null`	probe-model-probe-c1-pr-1.jsonl:2 «"observation": "c1.playback_session.initial_value"»
-	Change not in the PR description (probe model-probe-c1)	unmentioned change	c2.listener_callback_exists: `value`: false → true	probe-model-probe-c1-pr-1.jsonl:4 «"observation": "c2.listener_callback_exists"»

Citation check (PASS): 3 citation(s), every one re-read from its log file and matched.

What changed in behavior

What	Count	Examples
Tests that actually ran (main / PR, fewest of the runs)	1 / 6	-
Tests unchanged	1	-
New tests that pass	5	-
Tests that pass on main but fail on the PR	0	-
Tests that are new in the PR and fail	0	-
Tests that ran on main but are missing on the PR side	0	-
Tests that changed result in another way (for example passed → skipped)	0	-
Tests that gave different results between the 3 runs of the same side (flaky)	0	-
Tests that fail on main and pass on the PR	0	-
Tests that are new or changed but did not run because an optional module is missing	0	-
Tests that were re-parametrized (same test, new parameter ids; not counted as removed)	0	-

What	Count	Examples
Test modules that fail to load on the PR side only (their tests no longer run)	0	-
Test modules that fail to load on main only (no baseline for their tests)	0	-
Test modules that fail to load on both sides (their tests did not run)	0	-
Tests that fail identically on both sides (already broken on main)	0	-

Probes

Model-written probes: model provider: nebius (Token Factory, live).

A probe is a small deterministic script run on both sides in the same environments as the tests, 3 alternating runs per side. It prints JSON observations; plain code (poc/diff.py) compares them. A difference that no claim of the PR covers needs attention. Every probe was frozen (content sha256) before any PR-side code ran, and checked against that freeze before each PR-side run. Only a live model probe is blind; fixed probes and recorded model fixtures are known-answer cases.

model-probe-c1 (model (nvidia/nemotron-3-super-120b-a12b, live), `model-probes/model-probe-c1.py`, frozen `495fb5322484` at 2026-10-21T14:25:03.275776+00:00): MultizoneController.playback_session attribute

- P-C1 (PR description): The PR parses the playbackSession field from MULTIZONE_STATUS messages and exposes it as MultizoneController.playback_session, which is None when no session is attached and cleared on reset_members(). Expects the covered observations to differ.

Observation	Claim	Result	What the probe saw (main → PR)	Log lines (main, PR)
`c1.playback_session.exists`	P-C1	as claimed	`value`: false → true	probe-model-probe-c1-main-1.jsonl:1 «"observation": "c1.playback_session.exists"» probe-model-probe-c1-pr-1.jsonl:1 «"observation": "c1.playback_session.exists"»

Observation	Claim	Result	What the probe saw (main → PR)	Log lines (main, PR)
`c1.playback_session.initial_value`	P-C1	no change although the claim says it changes	same on both sides: `null`	probe-model-probe-c1-main-1.jsonl:2 «"observation": "c1.playback_session.initial_value"» probe-model-probe-c1-pr-1.jsonl:2 «"observation": "c1.playback_session.initial_value"»

Observation	Claim	Result	What the probe saw (main → PR)	Log lines (main, PR)
`c1.playback_session.value_after_reset`	P-C1	no change although the claim says it changes	same on both sides: `null`	probe-model-probe-c1-main-1.jsonl:3 «"observation": "c1.playback_session.value_after_reset"» probe-model-probe-c1-pr-1.jsonl:3 «"observation": "c1.playback_session.value_after_reset"»

Observation	Claim	Result	What the probe saw (main → PR)	Log lines (main, PR)
`c2.listener_callback_exists`	-	differs, not in the PR description	`value`: false → true	probe-model-probe-c1-main-1.jsonl:4 «"observation": "c2.listener_callback_exists"» probe-model-probe-c1-pr-1.jsonl:4 «"observation": "c2.listener_callback_exists"»

Observation	Claim	Result	What the probe saw (main → PR)	Log lines (main, PR)
`c2.playback_session_update_class_exists`	P-C1	as claimed	`value`: false → true	probe-model-probe-c1-main-1.jsonl:5 «"observation": "c2.playback_session_update_class_exists"» probe-model-probe-c1-pr-1.jsonl:5 «"observation": "c2.playback_session_update_class_exists"»

Observation	Claim	Result	What the probe saw (main → PR)	Log lines (main, PR)
`c3.device_id_normalization.first_stored_id`	-	same on both sides	same on both sides: `"ABCDEF123456"`	probe-model-probe-c1-main-1.jsonl:6 «"observation": "c3.device_id_normalization.first_stored_id"» probe-model-probe-c1-pr-1.jsonl:6 «"observation": "c3.device_id_normalization.first_stored_id"»

Observation	Claim	Result	What the probe saw (main → PR)	Log lines (main, PR)
`c3.device_id_normalization.stored_ids`	-	same on both sides	same on both sides: `["ABCDEF123456"]`	probe-model-probe-c1-main-1.jsonl:7 «"observation": "c3.device_id_normalization.stored_ids"» probe-model-probe-c1-pr-1.jsonl:7 «"observation": "c3.device_id_normalization.stored_ids"»

Observation	Claim	Result	What the probe saw (main → PR)	Log lines (main, PR)
`c3.receive_message_result`	-	same on both sides	same on both sides: `true`	probe-model-probe-c1-main-1.jsonl:8 «"observation": "c3.receive_message_result"» probe-model-probe-c1-pr-1.jsonl:8 «"observation": "c3.receive_message_result"»

Observation	Claim	Result	What the probe saw (main → PR)	Log lines (main, PR)
`c4.outbound_message_types.are_only_known`	-	same on both sides	same on both sides: `true`	probe-model-probe-c1-main-1.jsonl:9 «"observation": "c4.outbound_message_types.are_only_known"» probe-model-probe-c1-pr-1.jsonl:9 «"observation": "c4.outbound_message_types.are_only_known"»

Observation	Claim	Result	What the probe saw (main → PR)	Log lines (main, PR)
`c4.outbound_message_types.sent_types`	-	same on both sides	same on both sides: `["GET_STATUS", "GET_CASTING_GROUPS"]`	probe-model-probe-c1-main-1.jsonl:10 «"observation": "c4.outbound_message_types.sent_types"» probe-model-probe-c1-pr-1.jsonl:10 «"observation": "c4.outbound_message_types.sent_types"»

Citation check (PASS): 20 citation(s), every one re-read from its log file and matched.

Was the changed code exercised?

Coverage: MEASURED by coverage.py on the PR side (2.2 s run).

Coverage gate (python, confidence normal): 41 of 43 changed executable line(s) ran (95%).

Changed file	Changed executable lines	Of those, ran
`pychromecast/controllers/multizone.py`	43	41

How this is measured: one extra, untimed run of the PR side's tests under coverage.py. Changed lines come from comparing the tested main and PR files. Compiled code (C, C++, Cython) is invisible to coverage.py; such a change only counts as verified when the PR adds or changes a test for that module.

What went right

- Tested `e863b6dfeeb6` of the PR against main (source-pair route).
- 3 alternating runs per side: main 3, PR 3 of 3.
- pytest exit codes explained: all runs exited 0.
- Changed code was exercised (coverage gate): 41 of 43 changed executable line(s) ran (95%).
- Probes ran on both sides: 1 probe(s), 3 alternating runs per side, every run printed its observations.
- Single-thread environment: OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1, MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1, BLIS_NUM_THREADS=1, PYTHONHASHSEED=0.
- Machine type recorded: local x86_64 8 cpus.
- Nothing posted upstream: GitHub was only read.
- Every claim cites a real log line: 3 citation(s), every one re-read from its log file and matched.
- Every probe observation cites its output line: 20 citation(s), every one re-read from its log file and matched.

What went wrong and how to fix it

Issue	Status	Detail	How to fix
Behavior difference found	finding	probe: 3 observation(s) need attention: c1.playback_session.initial_value does not differ although the claim says it does (C1); c1.playback_session.value_after_reset does not differ although the claim says it does (C1); c2.listener_callback_exists differs and no claim covers it; probe: claim(s) C1 proven; probe: 5 other observation(s) identical on both sides; tests: 1 test(s) ran on main, 6 on the PR; 5 new test(s) pass, 1 unchanged	This is about the PR, not the run: read the failing tests and their log lines and the Probes section, then decide.

Evals and guardrails

Check	Result	Detail
3 alternating runs per side	PASS	main 3, PR 3 of 3
pytest exit codes explained	PASS	all runs exited 0

Check	Result	Detail
Changed code was exercised (coverage gate)	PASS	41 of 43 changed executable line(s) ran (95%)
Probes ran on both sides	PASS	1 probe(s), 3 alternating runs per side, every run printed its observations
Single-thread environment	PASS	OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1, MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1, BLIS_NUM_THREADS=1, PYTHONHASHSEED=0
Machine type recorded	PASS	local x86_64 8 cpus
Nothing posted upstream	PASS	GitHub was only read
Known-answer replay (fresh re-run)	SKIP	not replayed yet
Every claim cites a real log line	PASS	3 citation(s), every one re-read from its log file and matched
Every probe observation cites its output line	PASS	20 citation(s), every one re-read from its log file and matched

Checklist

☐ I read the verdict and the plain-English summary.
☐ Every 'false' or 'unmentioned change' row has a log line I can open.
☐ Nothing under 'What went wrong' is a run problem (those mean Re-run, not Reject).
☐ I recorded my decision (see below).

Decision recorded: none yet. To record one, run `make decide REPORT=./report/pychromecast-1300 DECISION=approve` (or `reject`, `approve_with_notes`, `re_run`).

Rerun command

```
make check PR=https://github.com/home-assistant-libs/pychromecast/pull/1300
```

Evidence links

- Pull request home-assistant-libs/pychromecast#1300
- record.json
- pr.diff
- setup-main.log
- setup-pr.log
- git.log

- pytest-main-1.log
- pytest-pr-1.log
- pytest-main-2.log
- pytest-pr-2.log
- pytest-main-3.log
- pytest-pr-3.log
- junit-main-1.xml
- junit-pr-1.xml
- coverage-pytest.log
- coverage-pr.json
- probe-model-probe-c1-main-1.jsonl
- probe-model-probe-c1-pr-1.jsonl
- main run 1 result: pytest-main-1.log:7 «1 passed in 0.88s»
- pr run 1 result: pytest-pr-1.log:7 «6 passed in 0.97s»

Glossary

Term	Meaning
main	The library as it is today, without the PR.
PR	Pull request: a proposed change to the library.
run	One full pass of the selected tests. We do 3 per side, alternating main and PR.
flaky	A test that gives different results when nothing changed; it is ignored as evidence.
log line	A line in a pytest-*.log file: the raw output the verdict is based on.
route	How the two sides were built: nightly+patch, source-build or source-pair (see docs/OPERATIONS.md).
changed lines covered	How many of the PR's changed, executable Python lines ran during the PR side's tests.
re-parametrized	A test whose parameter ids changed (renamed cases); the new ids are judged on their own.